

B5: Programming Language Processing

CS1101S: Programming Methodology

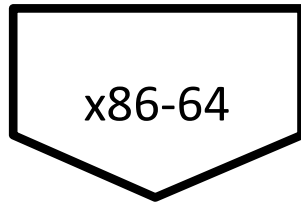
Sanka Rasnayaka

September 15, 2023

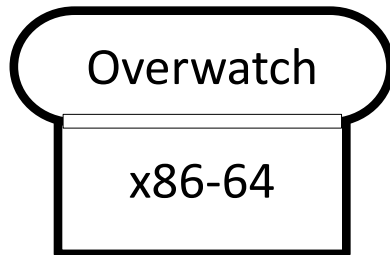
Outline

- T-Diagrams
- Interpreters
- Compilers
- Combinations
- Programming the LEGO Bricks

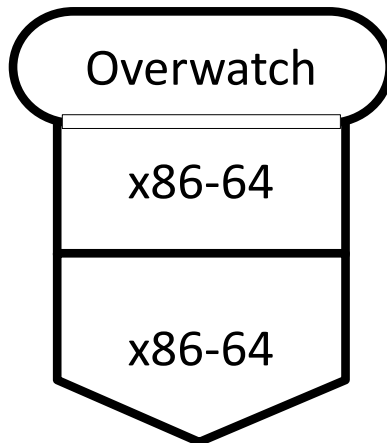
T-Diagrams (Tombstone Diagrams)



x86-64 Processor

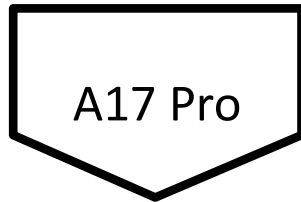


Program "Overwatch" (x86-64 code)

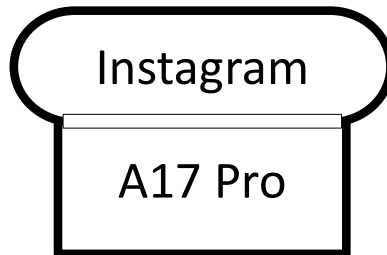


"Overwatch"
running on x86-64

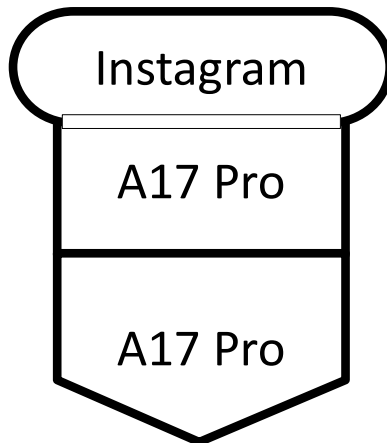
Running an app on iPhone 15



A17 Pro, the processor of iPhone 15



Program “Instagram” (A17 build)



“Instagram”
running on A17 Pro

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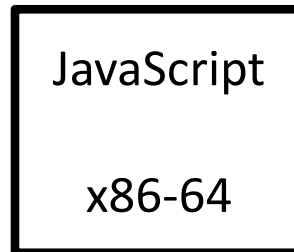
Interpreter

- Interpreter is a program that executes another program
- The interpreter's **source language** is the language in which the interpreter is written
- The interpreter's **target language** is the language in which the programs are written which the interpreter can execute

Teaser for Lecture 11

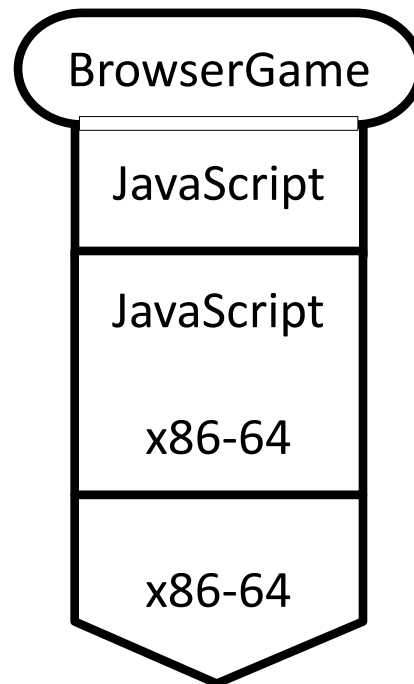
The evaluator (interpreter), which determines the meaning of statements in a language, is just another program. (“Most fundamental idea in programming”)

Interpreters



Chrome browser for PC, seen as an interpreter for JavaScript,
written in x86-64 machine code

Running JavaScript on Chrome



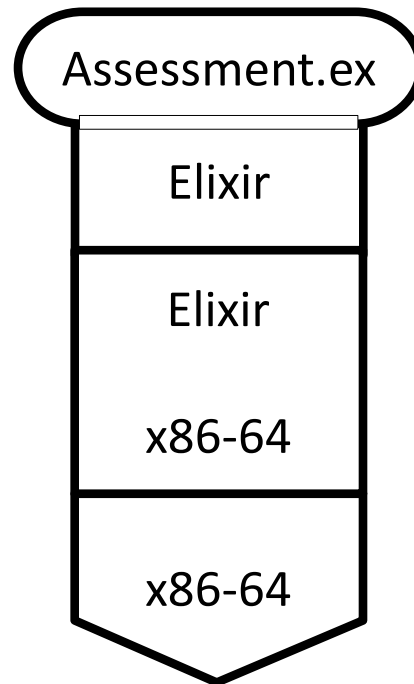
The browser acts as an interpreter for JavaScript.

Another example: Elixir



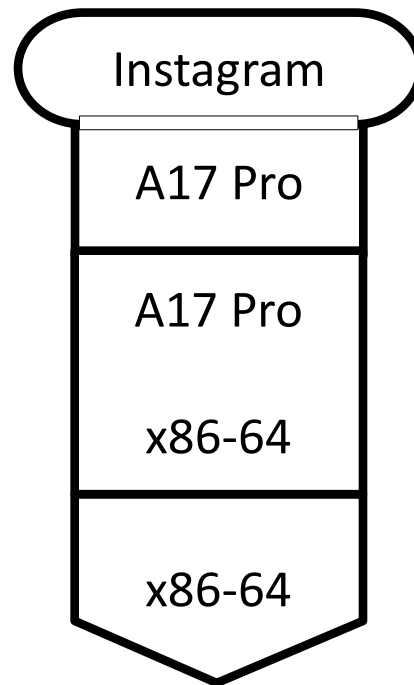
Interpreter for Elixir, written in x86-64 machine code

Running Elixir on Server



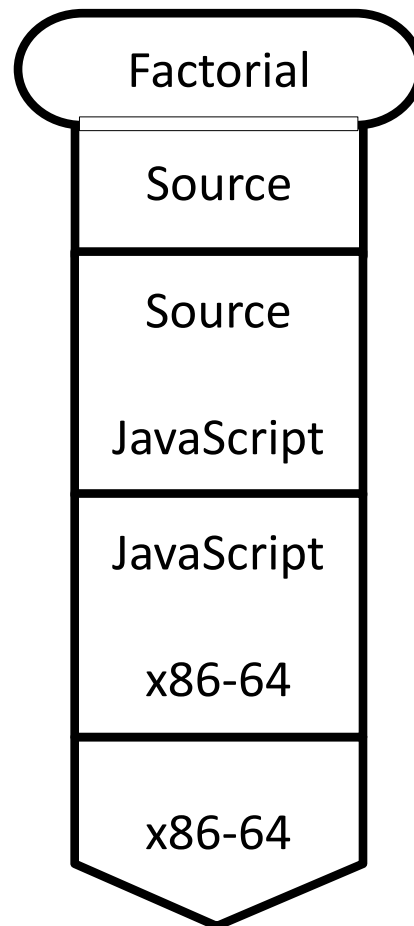
Elixir program “assessment” running on x86-64 using interpretation

Hardware Emulation



“Instagram” app running on a PC using hardware emulator

Running Source §2 in Source Academy using Stepper



Source Academy stepper:

layer between your programs and Chrome's native JavaScript

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Compilers

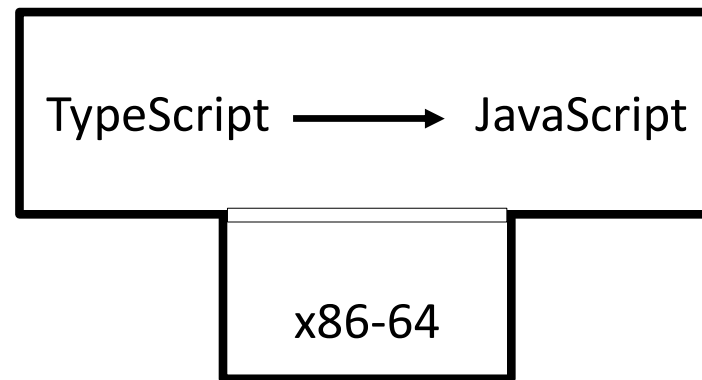
Definition:

A compiler is a program that translates from one language (the from-language) to another language (the to-language).

Teaser for Lecture 12C

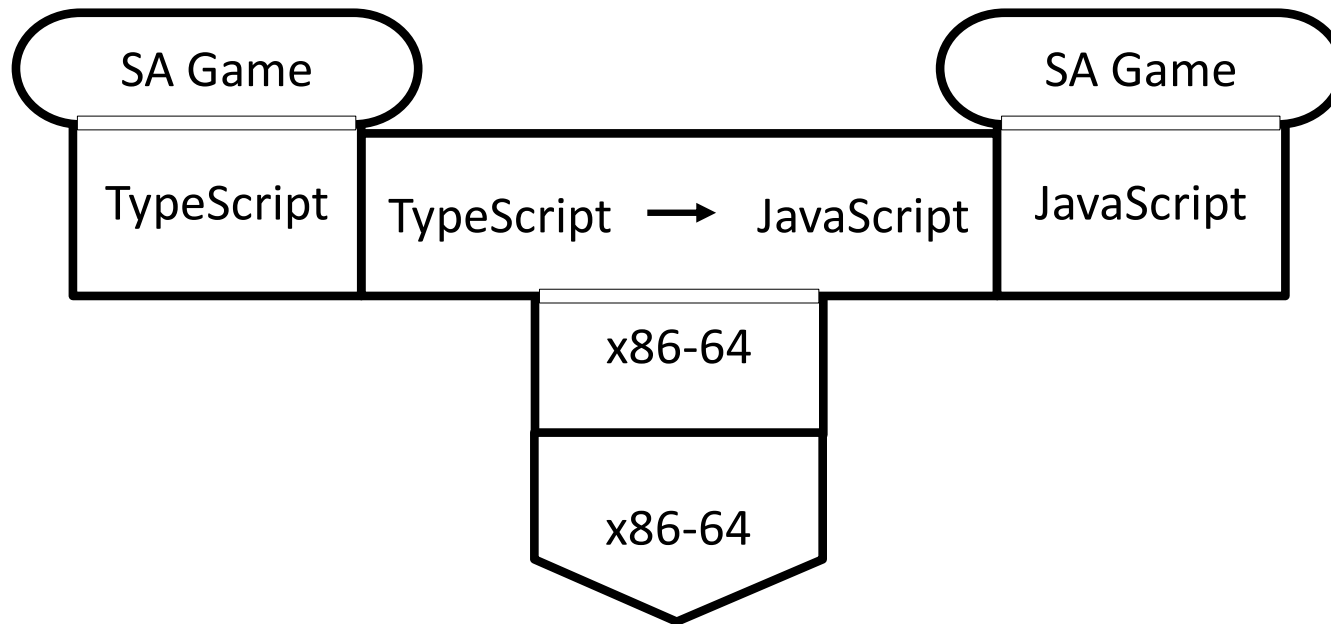
A compiler, which translates programs from one language to another, is just another program. (“Second most fundamental idea in programming”)

T-Diagram of Compiler



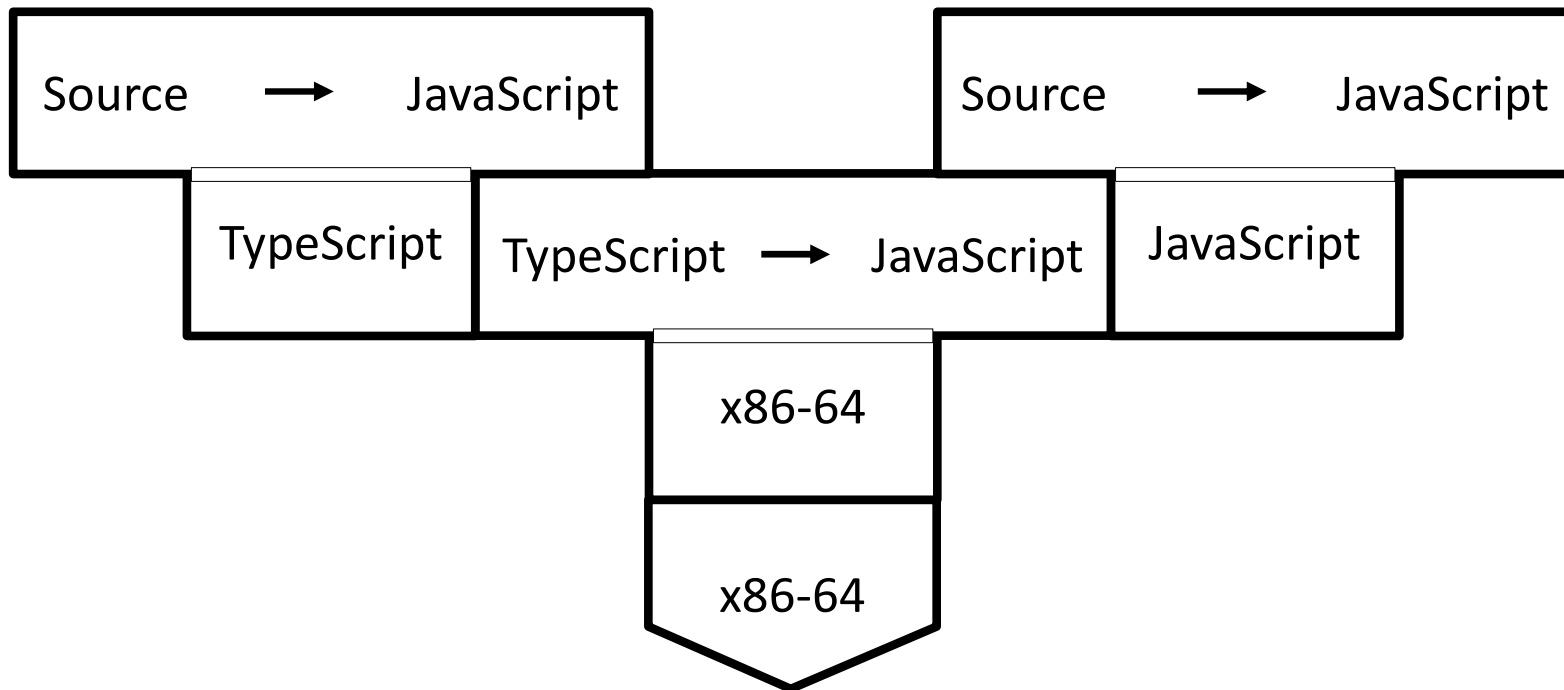
TypeScript-to-JavaScript compiler written in x86-64 machine code

Compiling a program (SA Game of Source Academy)



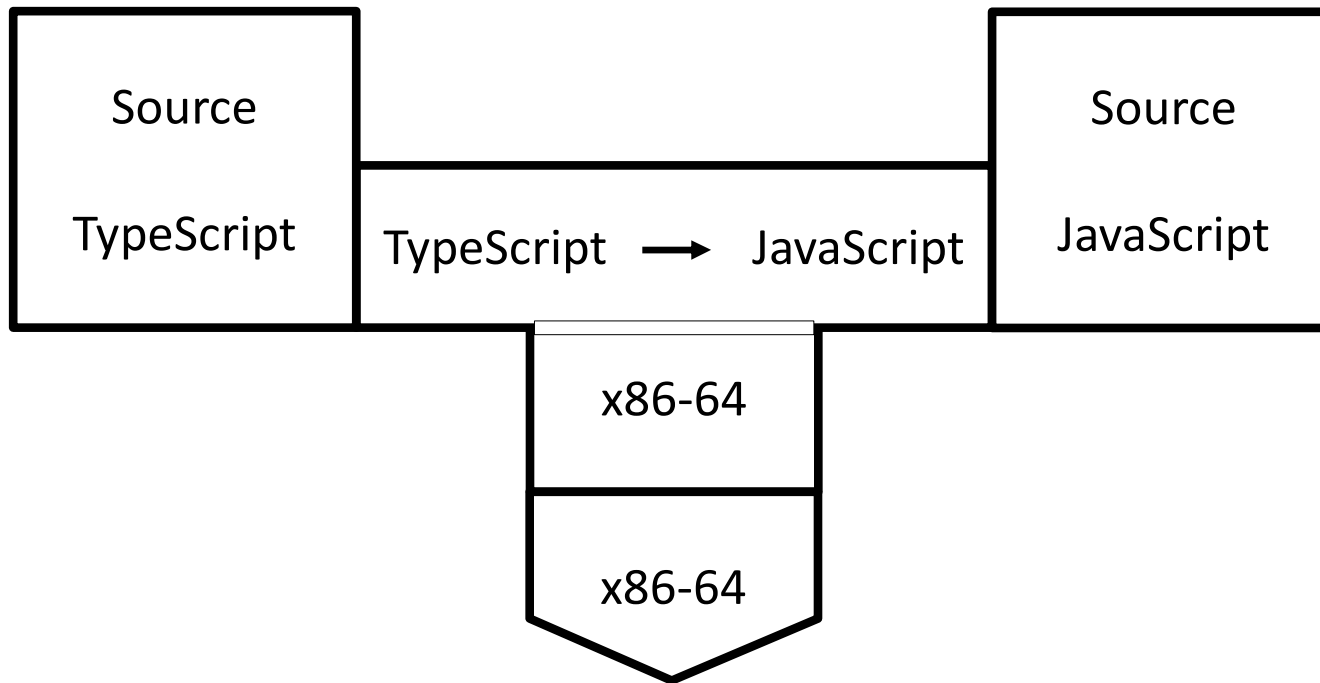
Compiling “SA Game” from TypeScript to JavaScript

Compiling our Source Compiler



Compiling Source-to-JavaScript compiler from TypeScript to JavaScript

Compiling the Stepper

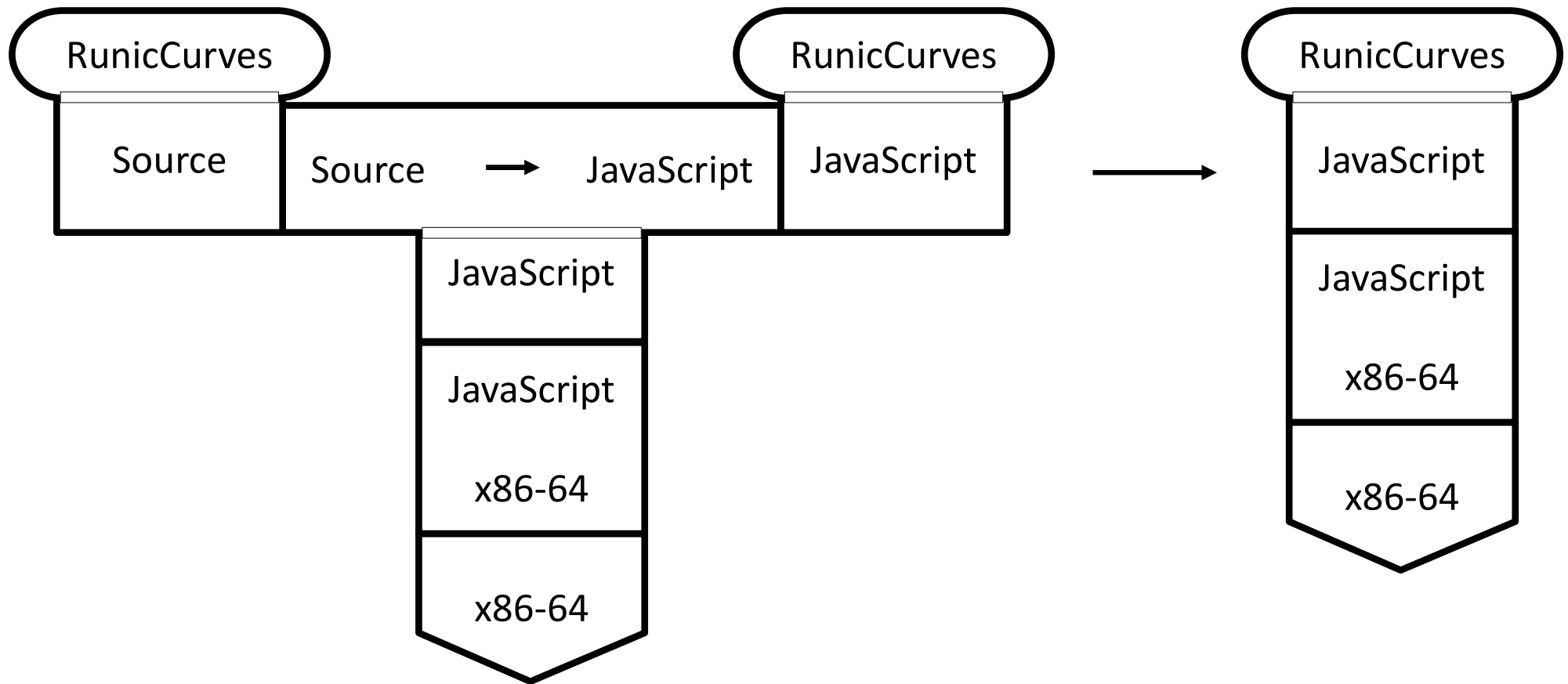


Compiling stepper tool from TypeScript to JavaScript

Outline

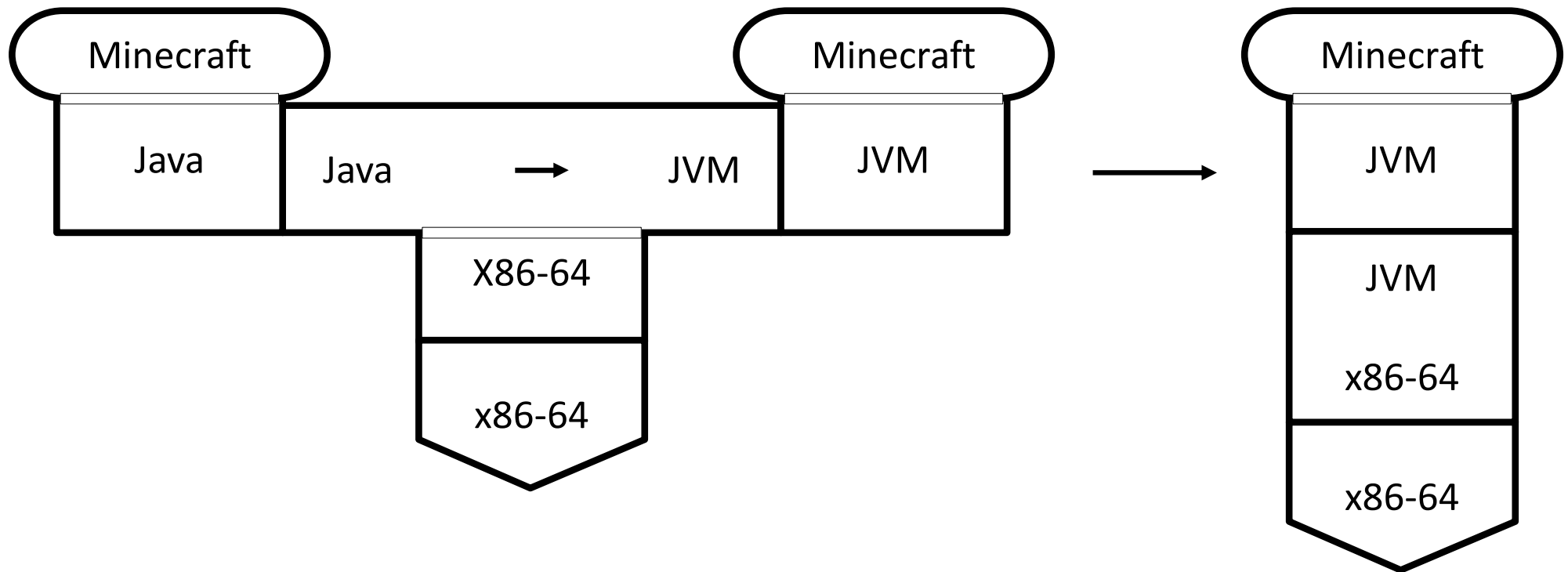
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A typical Source Academy session



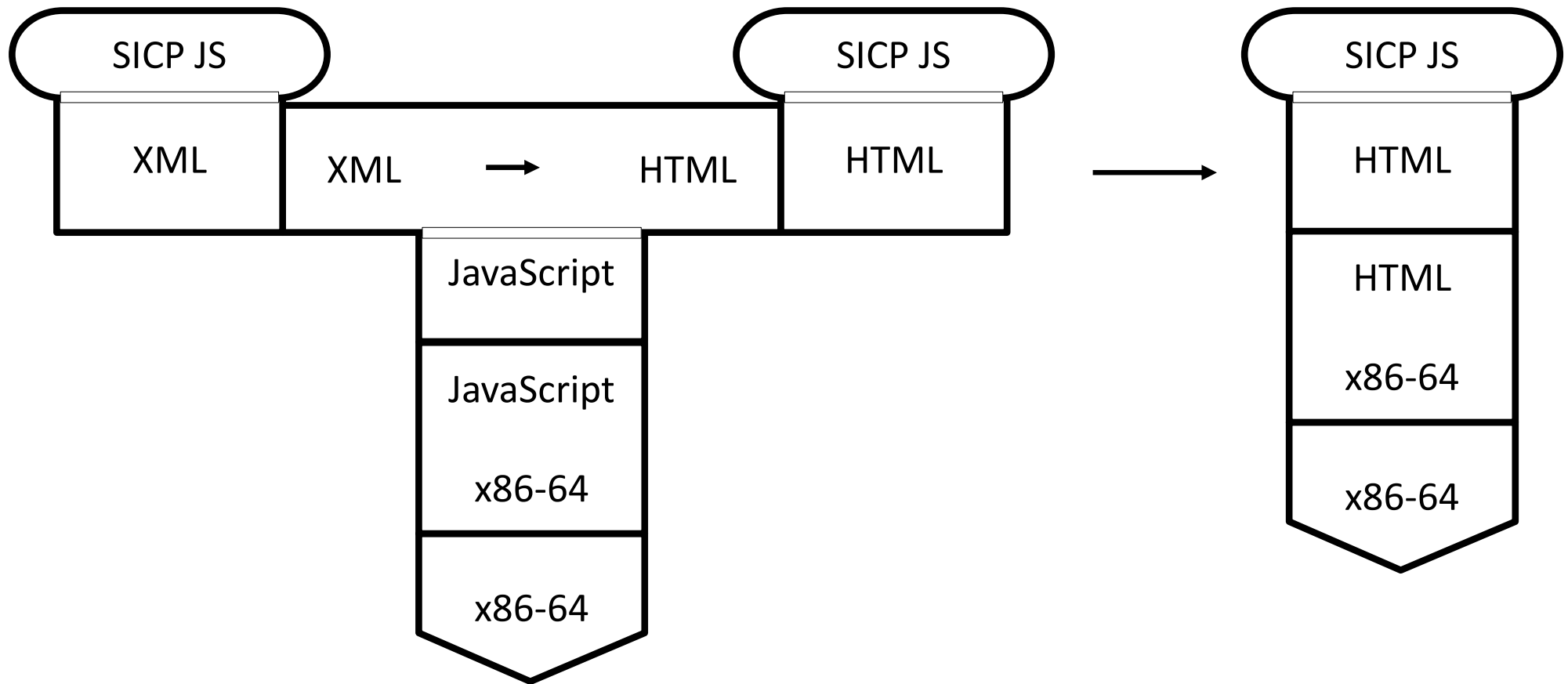
Compiling “RunicCurves” from Source to JavaScript in browser, and then running the JavaScript program in browser.

Typical Execution of Java Programs



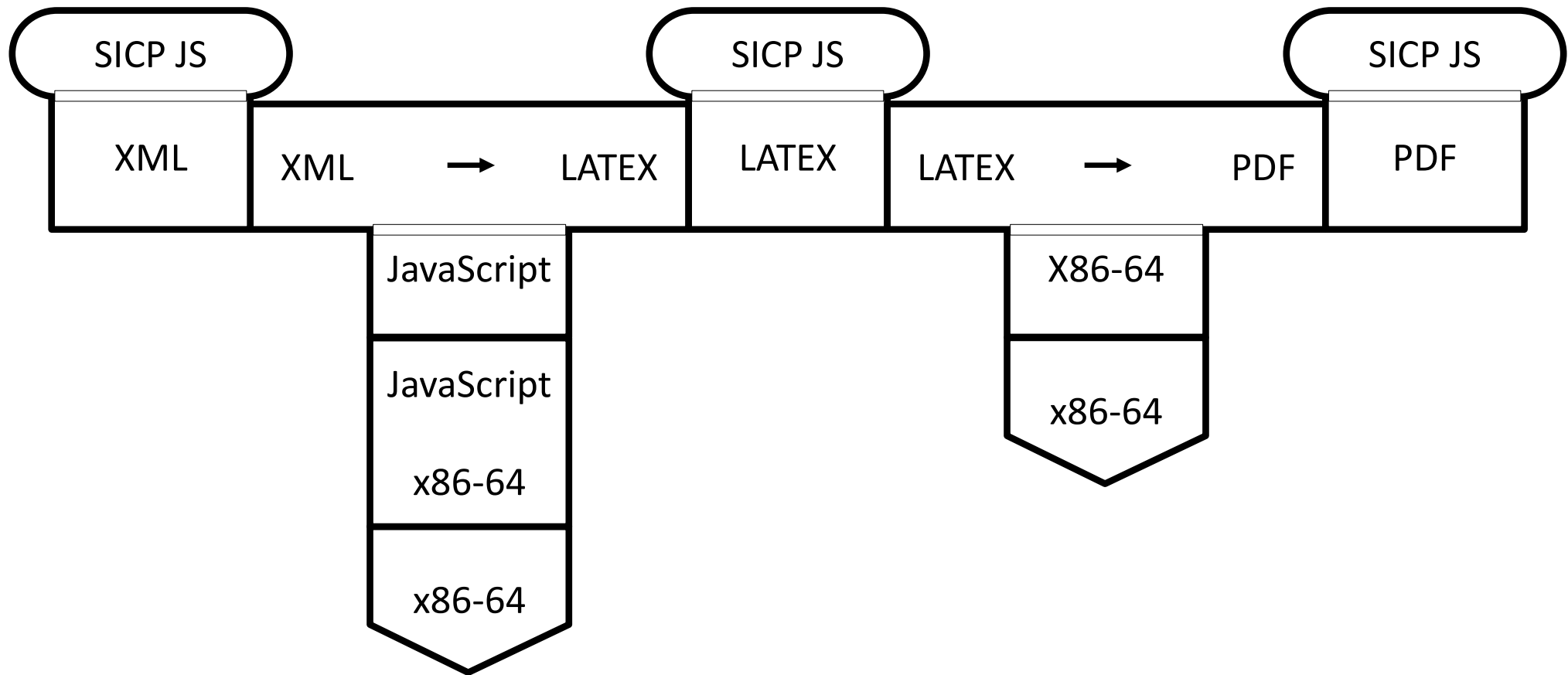
Compiling “RunicCurves” from Source to JavaScript in browser, and then running the JavaScript program in browser.

Excursion: Web edition of SICP JS



Compiling SICP JS textbook from XML to HTML, then viewing textbook with browser

Excursion: PDF edition of SICP JS

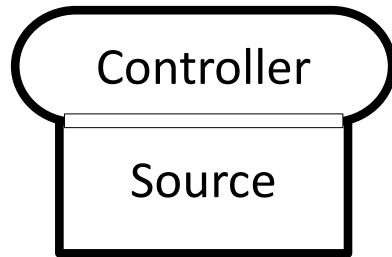


Compiling SICP JS from XML to PDF via LATEX

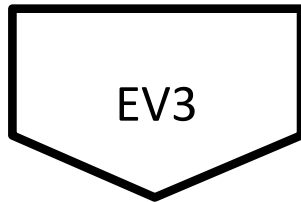
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Programming Lego Mindstorms with Source



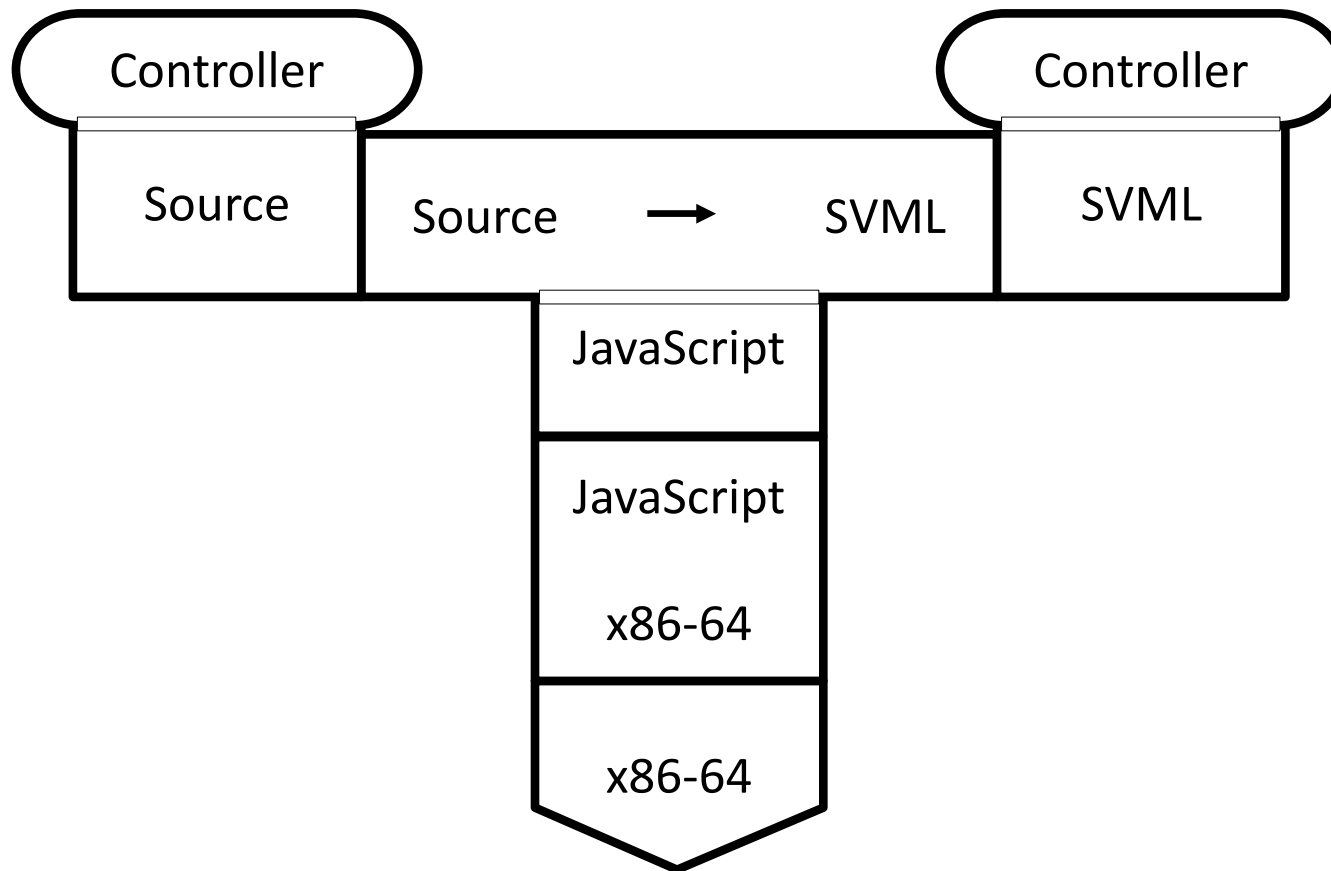
Program “Controller”
(written in Source)



Lego Mindstorms EV3 (ARM5 processor)
running EV3dev, an operating system
based on Debian Linux

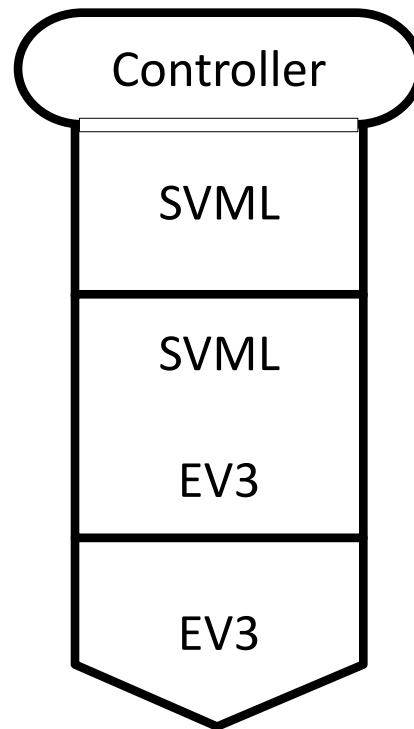
How to run Source on EV3?

SVML to the rescue!



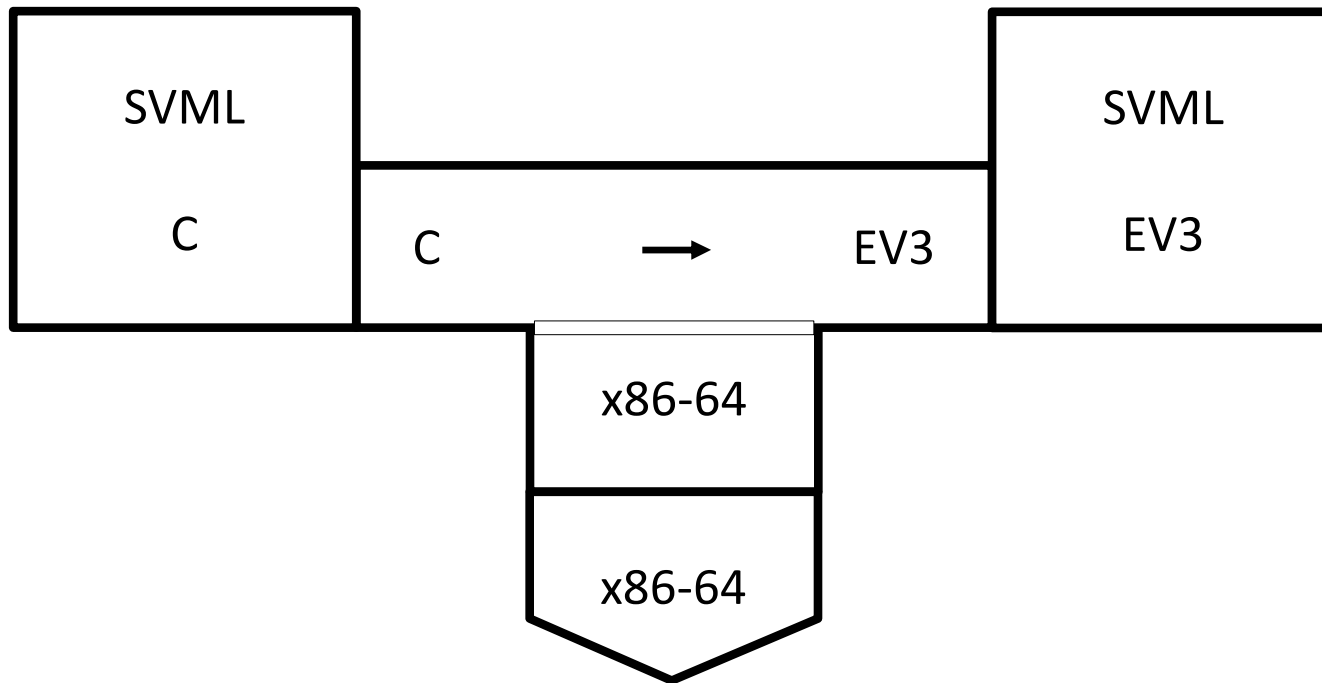
Compiling “controller” from Source to SVML

SVML on EV3



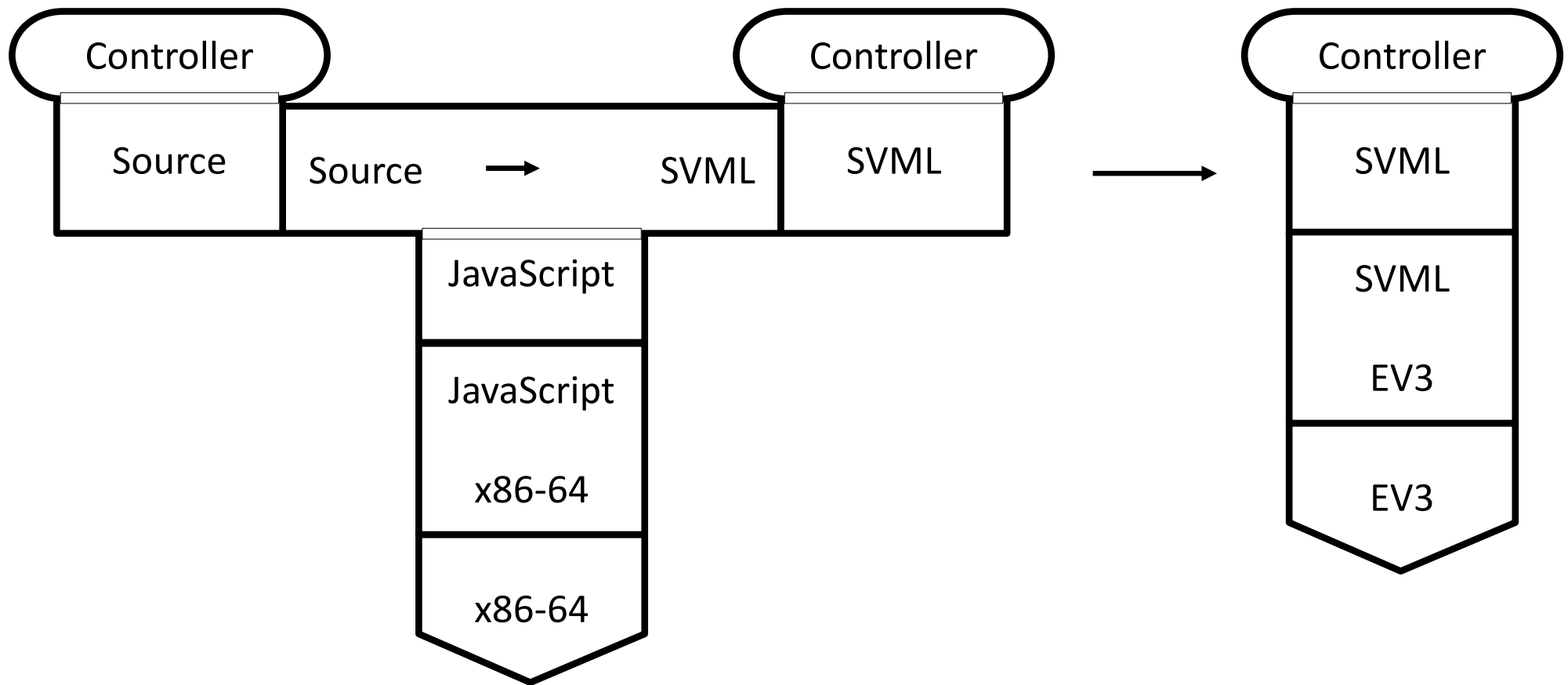
Running SVML program on EV3 brick

Compiling SVMML Emulator



Compiling SVMML emulator from C to EV3

Overview: Source for Robotics



Compiling “controller” from Source to SVML on a PC, and then running the SVML program on the EV3 brick.

Summary

- Components: programs, compilers, interpreters, machines
- T-diagrams
- Combinations of interpretation and compilation (tool chains) are common
- Source Academy is making use of Source interpreter and Source-to-JavaScript compiler, both written in TypeScript, compiled to JavaScript, and running on Chrome browser
- Robotics in CS1101S is making use of SVML interpreter and a Source-to-SVML compiler